# Technical Writing: From Documentation to Collaboration

## A Practical Overview of Modern Documentation Tools

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# Presentation Overview

1. Introduction
2. Documentation Tools
3. Version Control

# What is Technical Writing?

**Definition**

Technical writing is the process of communicating complex information in a clear, structured and understandable manner.

**Major characteristics:**

- Clarity and accuracy
- Logical organization
- Audience-focused communication
- Consistent formatting

# LaTeX: Structured Documentation

**LaTeX is used for:**

- Academic reports
- Research papers
- Mathematical documents
- Technical publications

**Example**

$$E = mc^2$$

Mathematical expressions can be professionally formatted.

# Beamer: Presentation Creation

Beamer is a LaTeX document class used to create structured presentations.

## Important Features

- Themes and color customization
- Columns and blocks
- Mathematical equations
- Tables, images and diagrams
- Automatic slide numbering

# Doxygen: Automated Code Documentation

Doxygen generates documentation directly from specially formatted source-code comments.

**Example Doxygen Comment**

```
@brief Calculates the average marks
@param marks Array of marks
@return Average value
```

**Applications:**

- Function documentation
- Class and module references
- HTML documentation generation

# Google Docs: Collaborative Writing

**Features**

- Real-time editing
- Comments and suggestions
- Version history
- Cloud storage

**Key Benefit**

Multiple users can contribute to the same document without maintaining separate copies.

# Git and GitHub

## Git

A distributed version control system used to track changes in files.

## GitHub

A platform for hosting Git repositories and collaborating on projects.

**Common Git workflow:**

Working Directory → Staging Area → Local Repository → GitHub

# Comparison of Documentation Tools

| Tool | Main Purpose | Output |
|---|---|---|
| LaTeX | Technical documents | PDF |
| Beamer | Presentations | PDF |
| Doxygen | Code documentation | HTML/PDF |
| Google Docs | Collaboration | Online documents |
| Git | Version control | Commit history |
| GitHub | Repository hosting | Online repository |

# Conclusion

**Key Takeaway**

Effective technical communication combines structured writing, clear presentation, automated documentation and version control.

Learn. Document. Collaborate. Improve.

# Thank You!

Questions and Discussion